A Face-Projection Spin(10) Effective Framework: Rubik Covariance, $O(2)$ Family Geometry, and the Conditional GUT Boundary

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(Dated: May 22, 2026)

Abstract

This draft rewrites the theory around a face-projection principle. Low-energy Standard-Model labels are not treated as primitive entries in a table; they are the projections of a higher-dimensional internal representation under a chosen symmetry-breaking face. In this language, one Spin(10) half-spinor 16 is the standard minimal six-face realization whose Standard-Model projection displays

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c.$$

Three families are protected separately by the index

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3.$$

The result is not an unconditional PSLT-only derivation of a unique grand unified theory. It is a conditional Spin(10) effective framework. The Majorana contact tensor direction is no longer left arbitrary: it is the unique $SL(2)$-invariant second-transvectant direction $K_{\rm tr}$. An optional Route-B hidden-messenger mechanism for $\zeta K_{\rm tr}$ is recorded as a conditional extension, but it is not part of the Route-A theorem core. An optional Route-D appendix records local string-constrained interpretations of the same skeleton. The hidden phase/radial sector and the companion audits for flavor, proton decay, thresholds, and source-sector UV origin remain open. Full phenomenological audits are deferred to companion work.

I. SCOPE OF THE LEAN ROUTE-A NOTE

This version is deliberately a Route-A paper skeleton. Its theorem-level claims are limited to three representation-geometric statements:

six faces	Spin(10) half-spinor restriction gives one SM family plus ν^c,
three copies	$\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ Dolbeault index gives three protected sections,
contact direction	$K_{\rm tr}$ is the unique $SL(2)$ second-transvectant direction.

Route B is kept only as a conditional candidate mechanism for the complex coefficient ζ. Route D is kept only as a local string-constrained interpretation of some of the same conditional data, not as a global compactification. Full CKM/PMNS fitting, dimension-five proton-decay Wilson tensors, threshold matching, and source-sector UV completion are not claimed in this lean note; they remain companion-audit problems.

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II. FACE PROJECTION

The organizing idea is that the visible particle table is a projection table, not a table of fundamental labels. Let G be a high-energy compact group, let R be a complex representation of G, and let a vacuum orientation θ select a broken subgroup

$$H_\theta \simeq \mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y.$$

The face projection is the restriction functor

$$\Pi_\theta(R) := \mathrm{Res}^G_{H_\theta} R = \bigoplus_\alpha m_{\alpha,\theta}\, r_{\alpha,\theta},$$

where the irreducible H_θ-summands $r_{\alpha,\theta}$ are the visible multiplet types. If $g \in G$, then the embedded subgroup changes by conjugation,

$$H_\theta \longmapsto H_{g\theta} = gH_\theta g^{-1}.$$

The same high-energy representation can therefore display different visible labels under a different face:

$$\Pi_{g\theta}(R) = \mathrm{Res}^G_{gH_\theta g^{-1}} R.$$

This is the representation-theoretic version of the intuitive statement that the same internal object can look different after a legal internal rotation.

Definition II.1 (Rubik covariance) *A unified model is face-covariant if the visible low-energy multiplet types are irreducible summands in restrictions of a common high-energy representation, and if every allowed change of face is a conjugation of the broken-subgroup embedding preserving anomaly cancellation, index, chirality, charge normalization, and representation dimension. The principle does not allow arbitrary transmutation of low-energy particles. It says that different low-energy labels can be different visible faces of the same high-energy object.*

The Rubik-cube image is useful only in this precise sense. A chemical element is still defined by nuclear charge; it does not become another element because one rotates a drawing. The analogy points instead to root lattices, Weyl chambers, gauge-basis rotations, and symmetry-breaking projections.

The concrete grand-unification branch used below follows the standard Pati–Salam and Spin(10) route [1–4]. The conditional neutrino sector uses the usual type-I seesaw logic [5–7], while the threshold language is the conventional perturbative RGE language [8, 9].

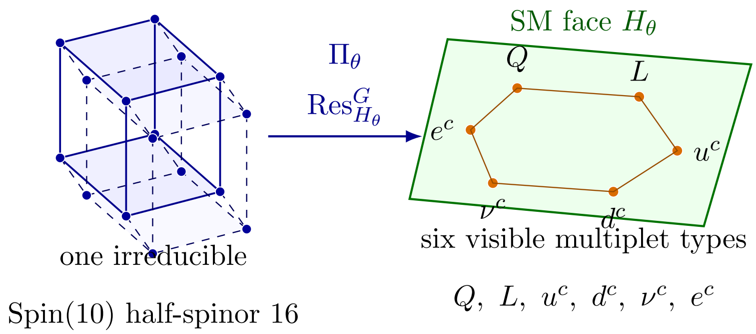

FIG. 1. Face-projection geometry. The 16 is visualized as a single higher-dimensional internal object whose Standard-Model face displays the six visible multiplet types $Q, L, u^c, d^c, \nu^c, e^c$. The figure is a representation-geometric visualization, not a dynamical particle-conversion process.

III. THE SIX-FACE Spin(10) OBJECT

Proposition III.1 (Standard minimal six-face realization) *Among standard simple compact GUT candidates, if the six chiral Standard-Model multiplet types*

$$Q, \quad L, \quad u^c, \quad d^c, \quad \nu^c, \quad e^c$$

are required to arise from one irreducible complex chiral representation whose restriction contains exactly one SM family plus ν^c, with no chiral SM exotics and with minimal representation dimension, the standard minimal single-object realization is the Spin(10) *half-spinor* 16.

Proof. The Pati–Salam subgroup of Spin(10) gives

$$16 \to (4, 2, 1) \oplus (\bar{4}, 1, 2).$$

Breaking

$$\mathrm{SU}(4)_C \to \mathrm{SU}(3)_C \times \mathrm{U}(1)_{B-L}$$

gives

$$4 \to (3, 1/3) \oplus (1, -1), \qquad \bar{4} \to (\bar{3}, -1/3) \oplus (1, +1).$$

With

$$Y = T_{3R} + \frac{B-L}{2},$$

the restriction to the Standard-Model face is

$$\mathrm{Res}^{\mathrm{Spin}(10)}_{G_{\mathrm{SM}}} 16 = Q \oplus L \oplus u^c \oplus d^c \oplus \nu^c \oplus e^c,$$

where the six projected Standard-Model multiplet types are

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$
Q	$(3,2)_{1/6}$
L	$(1,2)_{-1/2}$
u^c	$(\bar{3},1)_{-2/3}$
d^c	$(\bar{3},1)_{1/3}$
ν^c	$(1,1)_0$
e^c	$(1,1)_1$.

The $SU(5)$ alternative uses $10 \oplus \bar{5} \oplus 1$, which is not one irreducible chiral object. Pati–Salam is natural but not simple. E_6 contains this structure inside a larger 27, usually bringing extra matter that must be lifted. Therefore Spin(10) with a 16 is the standard minimal single-object realization. □

Proposition III.2 (Hypercharge normalization) *The face projection*

$$Y = T_{3R} + \frac{B-L}{2}$$

has the GUT normalization

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{5}{3}.$$

Proof. The trace is over one left-handed 16 after Standard-Model decomposition. For one family,

$$\mathrm{Tr}\, Y^2 = 6\left(\frac{1}{6}\right)^2 + 2\left(-\frac{1}{2}\right)^2 + 3\left(-\frac{2}{3}\right)^2 + 3\left(\frac{1}{3}\right)^2 + 1(1)^2 + 1(0)^2 = \frac{10}{3}.$$

The $SU(2)_L$ doublets are Q and L, and the doublet normalization is $\mathrm{Tr}_{\mathbf{2}}\, T_3^2 = 1/2$. Hence

$$\mathrm{Tr}\, T_{3L}^2 = 3 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = 2.$$

Thus

$$\frac{\mathrm{Tr}\, Y^2}{\mathrm{Tr}\, T_{3L}^2} = \frac{10/3}{2} = \frac{5}{3}.$$

□

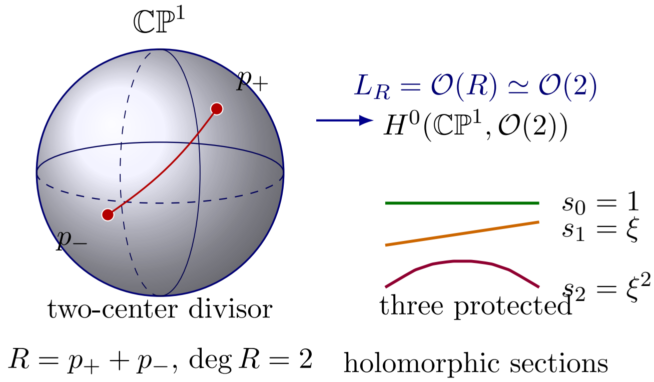

FIG. 2. Two-center geometry of the family carrier. The two marked points define a degree-two divisor $R = p_+ + p_-$, hence the line bundle $L_R \simeq \mathcal{O}(2)$. The family count comes from the protected section space $H^0(\mathbb{CP}^1, \mathcal{O}(2)) = \mathrm{span}\{1, \xi, \xi^2\}$, not from treating the two centers as two families. This is a topology/complex-geometry picture, not a dynamical flow diagram.

Remark III.3 (Allowed meaning of rotation) *At low energy Q cannot be freely rotated into e^c. The Standard-Model gauge symmetry has already selected fixed visible labels. The face-projection statement is about the high-energy representation and its broken-subgroup projection. Any physical process that changes the visible face must be mediated by legitimate high-energy fields and is constrained, for example, by proton decay.*

IV. THREE FAMILIES FROM AN $O(2)$ INDEX

The family number is not produced by rotating the six-face object. It is a separate index statement. The analytic input is the standard Spin^c Dolbeault–Dirac/cohomology identification and the Riemann–Roch–Serre-duality calculation on $\mathbb{CP}^1$ [10, 11].

Figure 2 is the geometric picture behind the theorem: the two centers set the degree of the line bundle, while Riemann–Roch and Serre duality fix the number of protected sections.

Theorem IV.1 (Conditional Spin^c index theorem for three protected families) *Assume that the protected internal family carrier is the* Spin^c *Dolbeault bundle on*

$$C = \mathbb{CP}^1$$

twisted by a two-center divisor

$$R = p_+ + p_-, \qquad L_R = \mathcal{O}(R) \simeq \mathcal{O}(2).$$

Assume also that all other SM-charged internal sectors are absent, gapped, or vectorlike, so they contribute no additional unpaired chiral zero modes. Let

$$D_R = \sqrt{2}\left(\bar{\partial}_{L_R} + \bar{\partial}^\dagger_{L_R}\right)$$

be the corresponding Spin^c *Dolbeault–Dirac operator. Then*

$$\ker D_R^+ \simeq H^0(\mathbb{CP}^1, \mathcal{O}(2)), \qquad \ker D_R^- \simeq H^1(\mathbb{CP}^1, \mathcal{O}(2)),$$

with

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3,$$

and $H^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0$. *Thus the protected sector contributes exactly three positive-chirality family zero modes.*

Proof. Hodge theory identifies the positive and negative Spin^c Dolbeault–Dirac kernels with the Dolbeault cohomology groups displayed above. Riemann–Roch on $\mathbb{CP}^1$ gives

$$h^0(\mathbb{CP}^1, \mathcal{O}(2)) - h^1(\mathbb{CP}^1, \mathcal{O}(2)) = 2 + 1 = 3.$$

Serre duality gives

$$h^1(\mathbb{CP}^1, \mathcal{O}(2)) = h^0(\mathbb{CP}^1, K_{\mathbb{CP}^1} \otimes \mathcal{O}(-2)) = h^0(\mathbb{CP}^1, \mathcal{O}(-4)) = 0.$$

Therefore $h^0 = 3$. □

Corollary IV.2 (Projected chiral state space) *Under the six-face* $\mathrm{Spin}(10)$ *assumption and the* $O(2)$ *family carrier, the low-energy chiral state space is*

$$\mathcal{S}^{4D}_{\text{chiral}} = 16 \otimes H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq 16_1 \oplus 16_2 \oplus 16_3.$$

Remark IV.3 (Why this is still conditional) *This theorem does not derive* $C = \mathbb{CP}^1$, *the two-center divisor* R, *the* Spin^c *carrier, or the no-extra-unpaired-sector condition. Those are branch assumptions. The theorem says that once those inputs are chosen, the family count is protected and is not a tunable scan output.*

Remark IV.4 (Tangent-bundle reading of the same carrier) *On the rational family curve,*

$$T_{\mathbb{CP}^1} \simeq \mathcal{O}(2), \qquad H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}_2(\mathbb{C}).$$

Thus, after the branch choice $L_R \simeq \mathcal{O}(2)$, the protected three-section space can also be read as the holomorphic vector-field algebra of the rational curve. This does not derive the curve $C = \mathbb{CP}^1$ or the divisor input; it only shows that the $\mathcal{O}(2)$ carrier, the three protected modes, and the $SL(2)$ family covariance are three views of the same rational-curve geometry. In an affine coordinate ξ, the vector field $\xi\partial_\xi$ vanishes at 0 and ∞; in this reading the two marked points $p_\pm$ can be viewed as a Cartan choice on the rational curve, up to $PGL(2)$ conjugation.

Remark IV.5 (Genus cap for the tangent reading) *The tangent-bundle reading is special to the rational curve. For a compact Riemann surface Σ_g,*

$$h^0(\Sigma_g, T_{\Sigma_g}) = \begin{cases} 3, & g = 0, \\ 1, & g = 1, \\ 0, & g \geq 2. \end{cases}$$

Indeed, $T_{\Sigma_g} \simeq K_{\Sigma_g}^{-1}$ has degree $2 - 2g$; for $g = 0$ it is $\mathcal{O}(2)$, for $g = 1$ it is trivial, and for $g \geq 2$ it has negative degree. Thus, within this tangent interpretation, three protected holomorphic vector fields are the rational-curve maximum, not a generic curve feature.

V. CONDITIONAL EFT DATA

The face-projection and index principles supply the representational skeleton. They do not by themselves determine the full action. The current conditional branch retains the following action-level data:

A1 Spin(10) face-projection EFT with complete 16_i,
A2 $H^0(\mathbb{CP}^1, \mathcal{O}(2))$ protects three copies of the 16,
A3 type-I seesaw with a source-fixed Majorana contact,
A4 proton bounds and two-loop threshold matching,
A5 constrained/composite 54/210 source order parameters,
A6 auxiliary/composite conormal multipliers,
A7 a Schur lock with singlet auxiliaries and sterile completion.

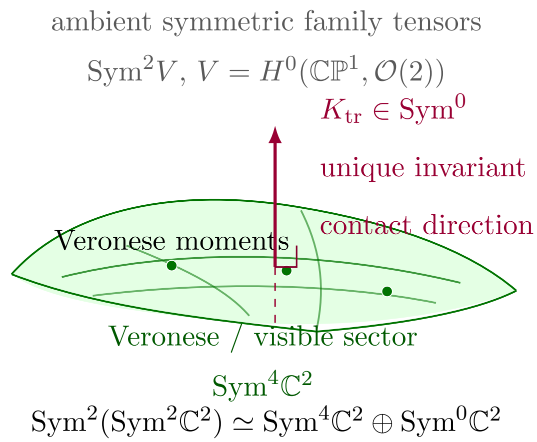

FIG. 3. Veronese sector and contact direction. The curved sheet visualizes the $\mathrm{Sym}^4\mathbb{C}^2$ Veronese/moment sector inside $\mathrm{Sym}^2 V$, while the transverse direction is the unique scalar summand $K_{\mathrm{tr}} \in \mathrm{Sym}^0$. The contact tensor is therefore not an arbitrary patch; it is the unique invariant direction outside the visible Veronese sector.

Figure 3 summarizes the representation-theoretic split used in the next lemma: the tensor direction of the Majorana contact is fixed, while its complex coefficient is separate conditional data.

Lemma V.1 (Unique $SL(2)$-invariant contact direction) *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathrm{Sym}^2\mathbb{C}^2.$$

Then

$$\mathrm{Sym}^2 V \simeq \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathrm{Sym}^0\mathbb{C}^2.$$

The Veronese moment sector is the $\mathrm{Sym}^4\mathbb{C}^2$ *summand, while the orthogonal singlet contact sector is unique up to phase. In the normalized* (ψ_0, ψ_1, ψ_2) *basis used for the family carrier, this singlet can be represented as*

$$K_{\mathrm{tr}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

Coordinate-free, $K_{\mathrm{tr}} \in \mathrm{Sym}^2 V^\vee$ *and its inverse bilinear form is* $K_{\mathrm{tr}}^{-1} \in \mathrm{Sym}^2 V$*. In the displayed orthonormal basis, the induced matrix obeys*

$$K_{\mathrm{tr}}^2 = \frac{1}{3} I, \qquad K_{\mathrm{tr}}^{-1} = 3K_{\mathrm{tr}}.$$

Proof. This is the Clebsch–Gordan decomposition for $\mathrm{Sym}^2(\mathrm{Sym}^2\mathbb{C}^2)$ [12]. The scalar summand has multiplicity one, so any invariant symmetric contact tensor is proportional to the displayed second-transvectant matrix. □

Remark V.2 (Killing-form reading) *Using the tangent-bundle identification*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq H^0(\mathbb{CP}^1, T_{\mathbb{CP}^1}) \simeq \mathfrak{sl}_2(\mathbb{C}),$$

the same invariant line can be read as the Killing-form line on the automorphism algebra of the rational family curve. The displayed K_{tr} differs from the conventional Killing matrix only by the chosen $SU(2)$-orthonormal basis, sign, and overall normalization. This gives the contact direction a geometric pairing interpretation, but it still does not turn $SL(2)$ into a fundamental global symmetry.

Remark V.3 (Meaning of the $SL(2)$ covariance) *The $SL(2)$ language in Lemma V.1 is used as family-geometry covariance, or equivalently as spurionic bookkeeping for the rational curve. It is not claimed to be an exact fundamental global symmetry. Any microscopic or quantum-gravity completion must realize this covariance geometrically, locally, or through spurions rather than as an unbroken global selection rule. Appendix B records one local string-constrained interpretation of this kind of geometric realization.*

Benchmark Datum V.4 (Majorana contact) *The benchmark Majorana matrix is not purely Veronese. It requires the unique orthogonal second-transvectant contact component*

$$\frac{M_R}{M_*} = M_V + \zeta K_{\mathrm{tr}},$$

with

$$\zeta = 0.1076472949 + 0.0736514853\, i.$$

The contact fraction is

$$\frac{\|\zeta K_{\mathrm{tr}}\|}{\|M_R/M_*\|} = 1.304275 \times 10^{-1}.$$

The companion machine audit that produced the previous long draft decomposes the inverse-seesaw reconstructed M_R into the Veronese subspace plus the orthogonal trace/contact direction. The Veronese-plus-contact residual is at machine precision, whereas the Veronese-only residual equals the displayed contact fraction. This means the contact is not a negligible correction to the present PMNS-compatible branch. The lean note records the datum and the geometrically fixed tensor direction; the full matrices, basis cards, and norm conventions are deferred to companion artifacts.

Observation V.5 (Hidden-origin scan) *Tested simple hidden-quotient and small monomial clockwork classes do not derive ζ. In those classes, it remains a conditional datum.*

The scan evidence is limited but sharp within that tested class. The contact-sensitivity scan gives the loose PMNS-compatible windows

$$\Delta s = 3.208798 \times 10^{-5}, \qquad \Delta\delta = 5.424119 \times 10^{-5} \text{ rad}.$$

Removing the contact spoils the PMNS and mass-splitting replay. A minimal hidden D-quotient leaves modulus and phase flat directions. A small real-coefficient monomial or clockwork phase with denominator $N \leq 6$ misses the target phase by at least

$$4.471595 \times 10^{-1} \text{ rad},$$

and the first root of unity inside the loose phase window appears only at

$$N = 178.$$

Thus no small hidden quotient or small monomial clockwork mechanism in the current evidence set predicts the complex value of ζ. A future non-minimal hidden sector may still do so, but it would be a new assumption that must be audited with its threshold and UV cost.

VI. OPTIONAL ROUTE B: HIDDEN TRANSVECTANT ORIGIN FOR ζ

The negative scan above does not forbid a hidden origin for ζ. It only says that the currently tested small hidden quotients and small monomial clockwork classes do not predict it. This section is an optional conditional extension, not part of the Route-A theorem core. The conservative Route-B move is therefore to add a new hidden-sector theorem: derive the tensor direction from the unique transvectant K_{tr}, and isolate the remaining burden in a hidden phase/radial sector that supplies the complex number.

Theorem VI.1 (Post-$B-L$ sterile transvectant messenger) *Let*

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)).$$

After $B-L$ breaking, let

$$N_i = P_{\nu^c}(16_i), \qquad N \in V,$$

be the right-handed neutrino family vector selected by the Majorana source projector. Let

$$X \in V^\vee$$

be a visible-singlet sterile family messenger. At the post-$B-L$ matching scale, consider the coordinate-free superpotential

$$\frac{W_R}{M_*} = \frac{1}{2} M_V(N, N) + \lambda X(N) - \frac{1}{2} K_{\mathrm{tr}}^{-1}(X, X),$$

where

$$M_V \in \mathrm{Sym}^2 V^\vee, \qquad K_{\mathrm{tr}} \in \mathrm{Sym}^2 V^\vee, \qquad K_{\mathrm{tr}}^{-1} \in \mathrm{Sym}^2 V.$$

In a chosen basis this is

$$\frac{W_R}{M_*} = \frac{1}{2} N^T M_V N + \lambda X^T N - \frac{1}{2} X^T K_{\mathrm{tr}}^{-1} X,$$

where K_{tr} is the unique $SL(2)$-invariant contact direction from Lemma V.1. Integrating out X gives

$$\frac{W_R^{\mathrm{eff}}}{M_*} = \frac{1}{2} \left(M_V + \lambda^2 K_{\mathrm{tr}} \right)(N, N).$$

Thus the benchmark contact is generated if

$$\lambda^2 = \zeta.$$

Proof. The X-F-term is the equality in V

$$\frac{\partial W_R}{\partial X} = M_* \left(\lambda N - K_{\mathrm{tr}}^{-1} X \right) = 0.$$

Here $K_{\mathrm{tr}}^{-1} : V^\vee \to V$ is the inverse map induced by the non-degenerate bilinear form K_{tr}. Thus

$$X = \lambda K_{\mathrm{tr}} N,$$

where $K_{\rm tr} : V \to V^\vee$. Substitution gives, in any basis,

$$\begin{aligned}\frac{W_R^{\rm eff}}{M_*} &= \frac{1}{2}M_V(N,N) + \lambda X(N) - \frac{1}{2}K_{\rm tr}^{-1}(X,X)\\ &= \frac{1}{2}M_V(N,N) + \lambda^2 K_{\rm tr}(N,N) - \frac{1}{2}\lambda^2 K_{\rm tr}(N,N)\\ &= \frac{1}{2}\left(M_V + \lambda^2 K_{\rm tr}\right)(N,N).\end{aligned}$$

Equivalently, in matrix notation,

$$\frac{W_R^{\rm eff}}{M_*} = \frac{1}{2}N^T M_V N + \frac{1}{2}\lambda^2 N^T K_{\rm tr} N.$$

This is exactly the displayed effective Majorana matrix. □

Remark VI.2 (Scope of the projector) *The field $N_i = P_{\nu^c}(16_i)$ is a post-$B-L$-breaking EFT variable. Before symmetry breaking, a* Spin(10)*-singlet messenger cannot select ν^c from the 16_i without an additional Majorana source or projector. Route B therefore assumes such a post-breaking projector, for example from a $(B-L) = -2$ Majorana source sector, rather than deriving it from face projection alone. In this lean note P_{ν^c} is therefore an input of the post-$B-L$ EFT matching problem, not a result of the face-projection principle. In conventional* SO(10) *language, one possible source of this post-breaking projection is a $\overline{126}_H$ Majorana source: its Pati–Salam $(10,1,3)$ component contains an SM-singlet direction that pairs with the $\nu^c\nu^c$ bilinear. This is only a source-sector interpretation here, not a completed UV model.*

Benchmark Datum VI.3 (Target coefficient) *For the current benchmark,*

$$\zeta_{\rm tar} = 0.1076472949 + 0.0736514853\, i.$$

Its polar data are

$$|\zeta_{\rm tar}| = 0.13043190325293763,$$

$$\arg \zeta_{\rm tar} = 0.600038020318215.$$

One square-root coupling is

$$\lambda = \sqrt{\zeta_{\rm tar}} = 0.34502115743308964 + 0.10673473744038917\, i,$$

with

$$|\lambda| = 0.3611535729477664.$$

This is a moderate hidden coupling; it is not a high-power clockwork suppression.

Remark VI.4 (Printed precision versus physical precision) *The many digits in* $\zeta_{\rm tar}$*,* $|\zeta_{\rm tar}|$*, and* λ *are kept as reproducibility anchors for the benchmark files. They should not be read as ten-digit physical predictions. The optional instanton interpretation in Appendix B already has expected higher-order corrections of order* $|\zeta|^2 \simeq 1.7 \times 10^{-2}$*, and the finite-messenger canonical-normalization estimate below gives* $|\lambda|^2/(16\pi^2) \simeq 8.26 \times 10^{-4}$*. Until the companion flavor/proton/threshold audits are replayed in a concrete UV completion, only the first few significant figures should be treated as physically meaningful.*

Observation VI.5 ($\mathbb{Z}_{178}$ phase diagnostic) *A discrete phase can approximate the benchmark phase by*

$$\zeta_{178} = |\zeta_{\rm tar}| \exp\left(\frac{2\pi i\, 17}{178}\right).$$

Since

$$\frac{2\pi\, 17}{178} = 0.6000794956295111,$$

the phase error is

$$4.1475311296 \times 10^{-5}\ \text{rad},$$

which lies inside the loose phase window

$$5.424119 \times 10^{-5}\ \text{rad}.$$

The direct complex error is

$$|\zeta_{178} - \zeta_{\rm tar}| = 5.4097037899 \times 10^{-6}.$$

This does not make ζ *an exact prediction. It supplies a precise discrete-phase candidate; exact equality would require either a small axion shift or a continuous hidden phase. The Route-B hidden-messenger theorem uses only* $\lambda^2 = \zeta$*; the* $\mathbb{Z}_{178}$ *value is kept as a side diagnostic, not as a premise of the benchmark branch. Diophantine approximations of this kind are generic for any real phase. For* $\arg \zeta/(2\pi) = 0.09549901697671904$*, the continued-fraction convergents include*

$$\frac{17}{178}, \qquad \frac{70}{733}, \qquad \frac{87}{911},$$

with phase errors

$$4.15 \times 10^{-5}, \qquad 6.68 \times 10^{-6}, \qquad 2.73 \times 10^{-6}\ \text{rad}.$$

Thus if the admissible phase window is tightened below the 178*-denominator error, the diagnostic must move to higher-denominator approximants. The stringy axion-volume option in Appendix B is the more physical way to interpret a continuous hidden phase.*

Observation VI.6 (Nonperturbative hidden-spurion option) *An alternative is*

$$\zeta = Ae^{-S_H + i\theta_H}.$$

For $A = 1$*,*

$$S_H = -\log|\zeta_{\mathrm{tar}}| = 2.0369040025655094, \qquad \theta_H = \arg \zeta_{\mathrm{tar}}.$$

This is naturally interpreted as a nonperturbative hidden-sector spurion. It is a conditional physical origin, not a weak-coupling derivation from face-projection alone. Appendix B records the local E3-instanton/Green–Schwarz version of this interpretation.

Lemma VI.7 (Minimal post-$B - L$ R-selection rule) *At the post-*$B - L$ *matching scale, take the superpotential to have* $R(W) = 2$*. A minimal selection rule that keeps the sterile messenger Majorana-only is*

object	*role*	$U(1)_R$
16_i *components*	*matter multiplets, including* N_i	1
$H, T, \bar{T}$	*Higgs doublet/triplet fields*	0
$X \in V^\vee$	*sterile transvectant messenger*	1
$\lambda,\ M_V,\ K_{\mathrm{tr}}^{-1}$	*matching tensors/spurions*	0

together with the post-breaking projector

$$N_i = P_{\nu^c}(16_i).$$

Then the Route-B Majorana terms

$$X(N), \qquad K_{\mathrm{tr}}^{-1}(X, X), \qquad M_V(N, N)$$

all have $R = 2$*, while ordinary Dirac and triplet Yukawa terms*

$$16_i 16_j H, \qquad 16_i 16_j T$$

also have $R = 2$*. However any superpotential operator with at least one additional* X *inserted into a Dirac or colored-triplet operator has*

$$R = 2 + m, \qquad m \geq 1,$$

and is therefore forbidden if the R*-selection rule is exact at the matching scale.*

Proof. The allowed Route-B terms have

$$R(X) + R(N) = 1 + 1 = 2, \qquad 2R(X) = 2, \qquad 2R(N) = 2.$$

The baseline Dirac and triplet Yukawa terms have

$$R(16_i) + R(16_j) + R(H \text{ or } T) = 1 + 1 + 0 = 2.$$

Adding $m \geq 1$ sterile messengers to such an operator shifts the superpotential charge to $2 + m$, so the operator is not allowed. This proves the selection rule at the holomorphic matching level. If later R-breaking spurions are introduced, they must be charged so that these forbidden operators remain absent to the order being audited. □

Remark VI.8 (Status of the R-selection rule) *The $U(1)_R$ assignment in Lemma VI.7 is used here as a holomorphic EFT matching selection rule. It is not yet claimed to be an anomaly-free microscopic symmetry. A UV implementation would require either an anomaly-free continuous or discrete R-symmetry, or a spurionic origin whose R-breaking insertions do not regenerate X-decorated Dirac or colored-triplet operators within the audited order. Appendix B gives a local Green–Schwarz/Stueckelberg interpretation of the same kind of selection-rule bookkeeping, again without claiming a global compactification.*

Theorem VI.9 (Tree-level matching silence) *In a canonical-normalized post-$B-L$ EFT at the matching scale, use Route B as a post-$B-L$-breaking matching theorem, with*

$$N_i = P_{\nu^c}(16_i)$$

as above. Assume the hidden messenger sector obeys the R-selection rule of Lemma VI.7:

$$X \text{ is a visible singlet}, \qquad X \text{ couples only to the Majorana channel } K_{\mathrm{tr}}(N, N),$$

and X has no couplings to Dirac Yukawa operators or colored-triplet operators. If integrating out X gives

$$\Delta M_R / M_* = \zeta K_{\mathrm{tr}},$$

then the visible one-loop threshold vector, the Dirac Yukawa matrices, and the dimension-five colored-triplet Wilson tensors are unchanged relative to the benchmark.

Proof. Since X is a visible singlet,

$$\Delta b_1^X = \Delta b_2^X = \Delta b_3^X = 0,$$

so it contributes no non-universal visible one-loop threshold. The Majorana-only selection rule gives

$$\delta Y_u = \delta Y_d = \delta Y_e = \delta Y_{\nu D} = 0.$$

The same rule forbids hidden insertions in the colored-triplet Yukawa couplings and inverse triplet propagator, so

$$\delta C_{5L} = \delta C_{5R} = 0.$$

If ζ equals the benchmark value, then M_R is the benchmark Majorana matrix and the type-I seesaw replay is unchanged. Thus the hidden contact can generate the missing Majorana tensor without moving the visible flavor, proton-decay, or one-loop threshold audits. □

Observation VI.10 (Canonical basis caveat) *The theorem above holds at tree level after canonical normalization. A microscopic completion with a finite messenger interval can generate a right-handed-neutrino wave-function correction of order*

$$\delta Z_N \sim \frac{|\lambda|^2}{16\pi^2} \log \frac{M_*}{M_X}.$$

For the benchmark value,

$$\frac{|\lambda|^2}{16\pi^2} = 8.259697 \times 10^{-4}.$$

This is small, but not automatically smaller than the narrow contact-sensitivity windows. Therefore the benchmark matrices in this lean note are defined after canonical normalization. If a microscopic model predicts a non-negligible δZ_N, the companion seesaw replay must use

$$Y_{\nu D} \mapsto Y_{\nu D} \left(1 - \frac{1}{2}\delta Z_N\right),$$

$$M_R \mapsto M_R - \frac{1}{2}\left(\delta Z_N^T M_R + M_R \delta Z_N\right),$$

before comparing to the PMNS benchmark.

The compact numerical checks for this optional Route-B extension are recorded separately in Appendix C; they are algebraic matching checks, not companion phenomenology audits.

VII. THEOREM BOUNDARY

Theorem VII.1 (Conditional benchmark boundary) *Given the face-projection data, the $O(2)$ index data, the companion benchmark audits, and either a benchmark-specified Majorana datum $\mathcal{D}_\zeta$ or a Route-B hidden messenger producing $\mathcal{D}_{\text{hidden zeta}}$, the current construction defines one internally checked conditional EFT branch:*

$$\frac{\mathcal{D}_{\text{face}} + \mathcal{D}_{\text{index}} + (\mathcal{D}_\zeta \textit{ or } \mathcal{D}_{\text{hidden zeta}}) + \mathcal{D}_{\text{companion audits}}}{\rightsquigarrow \;\; \textit{one conditional benchmark branch.}}$$

This does not support the stronger implication

$$\textit{PSLT/Another Physics alone} \Longrightarrow \textit{a unique GUT with predicted } \zeta.$$

Proof. A complete microscopic GUT requires an explicit gauge group, chiral matter representations, anomaly cancellation, a breaking potential, Yukawa and Majorana sectors, baryon-violating operators, and threshold matching. Face projection motivates the Spin(10) representation geometry. The $O(2)$ index protects the family count. Neither supplies all action-level data. The remaining branch data A1–A7 and the benchmark-level checks are conditional inputs or companion artifacts, and they are required whether ζ is supplied as a benchmark datum or generated through Route B. The Route-B messenger derives ζK_{tr} only after adding a hidden sterile messenger, phase/radial data, and Majorana-only selection rules. It therefore improves the conditional EFT explanation of ζ, but it is still not a PSLT-only prediction and it does not replace the companion phenomenological audits. □

Corollary VII.2 (Status of the rewritten theory) *The complete claim of this lean version is*

$$\boxed{\begin{array}{c}\textit{face-projection motivation} + \textit{ conditional } \mathrm{Spin}(10) \textit{ EFT} \\ \neq \textit{unconditional PSLT-only GUT proof.}\end{array}}$$

The rewrite is nevertheless useful: it gives a coherent representation-level reason why Spin(10) *is the natural minimal six-face family object and why three families are protected by an independent $O(2)$ index.*

VIII. MINIMAL REPRODUCIBILITY MANIFEST

The lean note has a small, explicit reproducibility surface. It does not include the deferred full flavor/proton/threshold scans.

- Paper source: `paper/gut_framework.tex` and `paper/refs.bib`.
- Compiled note: `paper/gut_framework.pdf`.
- D5 appendix check: `python3 code/verify_d5_half_spin_hypercube.py`.
- D5 outputs: summary, weights, and report in `output/d5_half_spin/`.
- Optional Route-B algebra check: `python3 code/verify_hidden_zeta_origin.py`.
- Optional Route-B output: verification JSON in `output/hidden_zeta_origin/routeB_hidden_zeta_verification.json`.
- Optional Route-D local string checks: `python3 route_d/code/verify_d1_f_theory_local_placement.py` and `python3 route_d/code/verify_d2_e3_instanton_zeta.py`.
- Optional Route-D outputs: `route_d/output/d1_f_theory_local_placement.*` and `route_d/output/d2_e3_instanton_zeta.*`.
- Deferred-audit convention card: `code/audit0_conventions_card.py`, with local output `output/audit0/invariant_card.*`.
- Deferred-audit scaffolds: `code/audit1_flavor_target_conventions.py` and `code/audit4a_source_spectrum_schema.py`, with local outputs in `output/audit1/` and `output/audit4a/`.
- Status log: `roadmap.md`.

The reproducible numerical claims inside this TeX draft are therefore only the D5 half-spin branching check, the hypercharge normalization check, the Route-B algebraic $\zeta K_{\rm tr}$ matching check, and the appendix-level Route-D local string-placement/instanton arithmetic. The Audit 0/1/4a files are convention and schema cards for future work, not phenomenology results. All publication-grade phenomenology remains outside this manifest until the companion audits are restored.

IX. DEFERRED CALCULATIONS

Earlier drafts contained detailed scan and audit material for Yukawa textures, $d = 5$ proton decay, threshold corrections, source sectors, clockwork rescue, and theorem ledgers. Those artifacts are preserved in the project history, but they are omitted from this lean TeX draft because their status relative to the face-projection principle is not yet settled. The only numerical Route-B checks kept here are the algebraic ζ, $K_{\rm tr}$, and phase-spurion checks needed for the optional hidden messenger theorem.

The paper strategy is therefore:

Route A	main paper: representation/index/contact skeleton,
Route B	optional extension or companion: hidden origin for ζ,
Route D	optional appendix: local string-constrained interpretation,
companion audits	full flavor, $d = 5$ proton decay, thresholds, source-sector UV.

The companion-audit dependency graph is fixed in the generated Audit 0 card as

$$0 \longrightarrow (1 \parallel 4a) \longrightarrow 3 \longrightarrow 2,$$

with $4b$ as an optional Route-D/global-geometry track. Audit 1 currently fixes only the flavor/seesaw target-table contract and marks publication targets as requiring refresh. Audit 4a currently fixes only the source-sector heavy-spectrum schema and records that no non-placeholder `heavy_spectrum.json` has yet been exported. Thus Audit 3 remains blocked on a source spectrum, and Audit 2 remains blocked on both physical flavor rotations and triplet inverse data.

X. LEAN-NOTE STATUS LEDGER

The logical status of this version is deliberately narrower than a complete GUT construction:

Proved/constructed.: Spin(10) face skeleton; $\mathrm{Spin}^c(\mathbb{CP}^1, \mathcal{O}(2))$ index; $K_{\rm tr}$ uniqueness; optional Route-B Schur complement.

Conditional inputs.: $C = \mathbb{CP}^1$, $R = p_+ + p_-$; protected Spin^c carrier; no unpaired SM-charged exotics; P_{ν^c}; R-selection rule; hidden phase/radial sector.

Deferred audits.: Full flavor fit; $d = 5$ proton-decay Wilson tensors; threshold/proton companion audits; source-sector UV origin.

Optional UV interpretations.: Route-D local F-theory placement and E3-instanton interpretation of ζK_{tr}, both with global compactification and zero-mode caveats.

Not claimed.: PSLT-only unique GUT proof.

XI. CONCLUSION

The clean result of the rewrite is a bounded theorem stack. A Spin(10) half-spinor supplies a standard minimal six-face object; the Spin$^c(\mathbb{CP}^1, \mathcal{O}(2))$ carrier supplies three protected copies; and the second-transvectant singlet supplies the unique Majorana contact direction. These are mathematical inputs and consequences inside a conditional EFT branch. They do not by themselves determine the microscopic source sector, the complex hidden coefficient ζ, the full flavor fit, or the proton-decay Wilson tensors. The optional Route-D appendix gives local string-constrained interpretations of some conditional inputs, not a global string compactification. Keeping this boundary explicit is the point of the lean Route-A version. This note establishes the representation/index/contact skeleton; it does not claim full GUT phenomenology.

Appendix A: D_5 Half-Spin Weight Hypercube

This appendix records the compact representation-theoretic calculation behind the six-face statement. It is not a new uniqueness theorem; it is the weight-level explanation of the standard Spin(10) half-spinor branch used in the main text [4, 12].

Let $e_1, \ldots, e_5$ be an orthonormal basis for the D_5 weight space. The roots of D_5 are

$$\pm e_a \pm e_b, \qquad a \neq b.$$

Choose the positive-chirality half-spin representation to have weights

$$\Lambda_{16} = \left\{ \frac{1}{2} \sum_{a=1}^{5} s_a e_a : s_a = \pm 1, \quad \#\{a : s_a = -1\} \equiv 0 \pmod 2 \right\}.$$

There are $2^{5-1} = 16$ such weights. The Pati–Salam face is the split

$$D_5 \supset D_3 \oplus D_2, \qquad \mathrm{Spin}(6) \times \mathrm{Spin}(4) \simeq \mathrm{SU}(4)_C \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R.$$

Use s_1, s_2, s_3 for the D_3 spinor weights and s_4, s_5 for the D_2 spinor weights. Define

$$m_C = \#\{a \le 3 : s_a = -1\}, \qquad m_W = \#\{a = 4, 5 : s_a = -1\}.$$

The chirality condition says $m_C + m_W$ is even, so m_C and m_W have the same parity. Therefore the weights split into exactly two blocks:

m_C parity	m_W parity	number of weights	Pati–Salam block
even	even	$4 \cdot 2 = 8$	$(4, 2, 1)$
odd	odd	$4 \cdot 2 = 8$	$(\bar{4}, 1, 2)$.

Thus

$$16 \to (4, 2, 1) \oplus (\bar{4}, 1, 2).$$

One convenient convention for the $\mathrm{SU}(4)_C$ component is

first three signs	$\mathrm{SU}(3)_C$	$B - L$
zero minus signs	1	-1
two minus signs	3	$1/3$
one minus sign	$\bar{3}$	$-1/3$
three minus signs	1	1.

For the weak D_2 signs, take

$$(s_4, s_5) = (+,+), (-,-) \quad \Longrightarrow \quad \mathrm{SU}(2)_L \text{ doublet}, \quad T_{3L} = \frac{s_4 + s_5}{4}, \quad T_{3R} = 0,$$

and

$$(s_4, s_5) = (+,-), (-,+) \quad \Longrightarrow \quad \mathrm{SU}(2)_R \text{ doublet}, \quad T_{3R} = \frac{s_4 - s_5}{4}, \quad T_{3L} = 0.$$

Applying

$$Y = T_{3R} + \frac{B - L}{2}$$

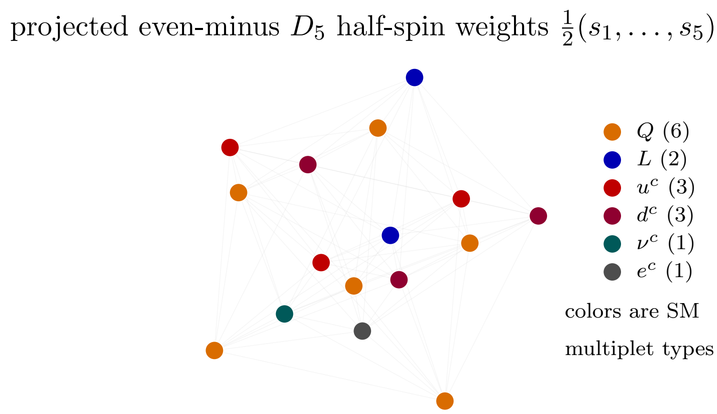

FIG. 4. Projected D_5 half-spin weight hypercube. The 16 even-parity half-spin weights are shown as one projected demihypercube and colored by their Standard-Model multiplet type after the Pati–Salam face projection. The multiplicities reproduce $(Q, L, u^c, d^c, \nu^c, e^c) = (6, 2, 3, 3, 1, 1)$.

then gives the six visible Standard-Model faces:

field	$\mathrm{SU}(3)_C \times \mathrm{SU}(2)_L$	Y	multiplicity
Q	$(3, 2)$	$1/6$	6
L	$(1, 2)$	$-1/2$	2
u^c	$(\bar{3}, 1)$	$-2/3$	3
d^c	$(\bar{3}, 1)$	$1/3$	3
ν^c	$(1, 1)$	0	1
e^c	$(1, 1)$	1	1.

The multiplicities sum to 16, so this is a complete half-spinor, not a collection of separately chosen low-energy fragments.

The appendix calculation is reproduced by

`code/verify_d5_half_spin_hypercube.py`.

The generated summary reports

$$\#\Lambda_{16} = 16, \qquad (4, 2, 1) : 8, \qquad (\bar{4}, 1, 2) : 8,$$

the six field multiplicities

$$(Q, L, u^c, d^c, \nu^c, e^c) = (6, 2, 3, 3, 1, 1),$$

and the same normalization check as in the main text,

$$\mathrm{Tr}\, Y^2 = \frac{10}{3}, \qquad \mathrm{Tr}\, T_{3L}^2 = 2, \qquad k_Y = \frac{5}{3}.$$

Appendix B: Optional Route-D String-Constrained Placement

This appendix records the promotion decision for Route D. The local F-theory placement audit and the E3-instanton audit are included only as a string-constrained interpretation of the Route-A/Route-B data. They are not part of the Route-A theorem core and do not constitute a global F-theory compactification.

1. Local $D_5 \to E_6$ placement

In local F-theory GUT models, a gauge algebra is engineered on a divisor by the singular fiber type. A D_5 singularity realizes $\mathfrak{so}(10)$, while charged matter can localize on curves where the singularity enhances in codimension two. The Katz–Vafa local rule identifies the matter from the adjoint branching of the enhanced algebra [13, 14]. For a $D_5 \to E_6$ enhancement,

$$E_6 \supset \mathrm{SO}(10) \times \mathrm{U}(1),$$

and

$$\mathbf{78} = \mathbf{45}_0 \oplus \mathbf{1}_0 \oplus \mathbf{16}_{-3} \oplus \overline{\mathbf{16}}_{+3}.$$

The dimension check is

$$45 + 1 + 16 + 16 = 78.$$

Thus the six-face Spin(10) object used in the main text has a standard local string interpretation: the **16** appears as one irreducible matter curve representation, rather than as six separately chosen Standard-Model fragments.

The same local picture can reproduce the $O(2)$ family carrier. On a matter curve $\Sigma = \mathbb{CP}^1$, the zero-mode carrier is represented as

$$K_\Sigma^{1/2} \otimes L_{\mathrm{flux}}.$$

Since

$$K_{\mathbb{CP}^1}^{1/2} = \mathcal{O}(-1),$$

a flux bundle with degree three gives

$$K_{\mathbb{CP}^1}^{1/2} \otimes \mathcal{O}(3) = \mathcal{O}(-1) \otimes \mathcal{O}(3) = \mathcal{O}(2).$$

Consequently

$$h^0(\mathbb{CP}^1, \mathcal{O}(2)) = 3, \qquad h^1(\mathbb{CP}^1, \mathcal{O}(2)) = 0.$$

This does not eliminate conditional input; it replaces the abstract $\mathcal{O}(2)$ assumption by the flux-quantized matter-curve datum

$$2 = 3_{\text{flux}} - 1_{\text{spin}}.$$

Remark B.1 (Global caveat) *The local $D_5 \to E_6$ branching and the $\mathbb{CP}^1$ index do not verify a global F-theory compactification. Hypercharge flux or Wilson-line breaking must still satisfy the global massless-hypercharge condition and avoid chiral exotics and uncontrolled threshold corrections [15, 16].*

2. E3-instanton interpretation of ζK_{tr}

The optional Route-B hidden messenger isolates the complex coefficient ζ from the tensor direction K_{tr}. A standard Type IIB/F theory interpretation of this coefficient is an Euclidean D3-brane instanton factor [17–20]

$$\zeta_\Gamma = A_\Gamma e^{2\pi i T_\Gamma}, \qquad T_\Gamma = \theta_\Gamma + i\, t_\Gamma.$$

Thus the hidden phase/radial sector becomes an axion-volume pair. For the benchmark value

$$\zeta_{\text{tar}} = 0.1076472949 + 0.0736514853\, i,$$

one has

$$|\zeta_{\text{tar}}| = 0.13043190325293763, \qquad \arg \zeta_{\text{tar}} = 0.600038020318215.$$

For unit prefactor modulus,

$$S_{\text{eff}} = -\log |\zeta_{\text{tar}}| = 2.0369040025655094,$$

or, in the convention $\zeta = \exp(2\pi i T_\Gamma)$,

$$\operatorname{Im} T_\Gamma = 0.3241833406119675.$$

The Majorana-only selection rule also has a standard instanton interpretation. Perturbative abelian symmetries can survive as selection rules even when the corresponding gauge bosons become massive by a Green–Schwarz/Stueckelberg mechanism. An instanton factor can transform under such a symmetry because its axion shifts. In a minimal charge ledger,

$$q(N) = +1, \qquad q(NN) = +2, \qquad q(e^{-S_\Gamma + i\varphi_\Gamma}) = -2,$$

so the instanton-dressed Majorana insertion is neutral:

$$q\big(e^{-S_\Gamma + i\varphi_\Gamma} NN\big) = 0.$$

In an actual compactification this bookkeeping must be realized by universal and charged instanton zero modes, whose saturation pattern inserts $N_i N_j$ and does not generate the forbidden Dirac or colored-triplet operators [21, 22].

Finally, landing specifically on $K_{\rm tr}$ requires the same locality or equivariance assumption used in the Route-B discussion. If the instanton data are blind to detailed family coordinates on

$$V = H^0(\mathbb{CP}^1, \mathcal{O}(2)) \simeq \mathrm{Sym}^2\mathbb{C}^2$$

except through the contact pairing, then the unique invariant contact direction is the scalar second-transvectant summand

$$K_{\rm tr} \in \mathrm{Sym}^0 \subset \mathrm{Sym}^2(\mathrm{Sym}^2\mathbb{C}^2) = \mathrm{Sym}^4\mathbb{C}^2 \oplus \mathrm{Sym}^0.$$

Remark B.2 (Precision caveat) *The instanton interpretation explains the form, selection rule, and moderate scale of ζ. It does not predict the benchmark value to many digits. For the benchmark,*

$$|\zeta|^2 = 0.01701248138618368,$$

which is about

$$\frac{|\zeta|^2}{3.208798 \times 10^{-5}} = 530.1823731560442$$

times the recorded contact-sensitivity window. Multi-instanton, multi-cover, prefactor, and moduli-stabilization effects therefore cannot be ignored in any claim of high-precision prediction.

The appendix-level checks are reproduced by

`route_d/code/verify_d1_f_theory_local_placement.py`

and

`route_d/code/verify_d2_e3_instanton_zeta.py`.

check	**value**	**role**
$\lvert\zeta_{\rm tar}\rvert$	0.13043190325293763	target modulus
$\arg\zeta_{\rm tar}$	0.600038020318215	target phase
$\sqrt{\zeta_{\rm tar}}$	$0.345021 + 0.106735\,i$	hidden messenger coupling
$\lvert\lambda\rvert$	0.3611535729477664	moderate coupling
$\lVert\lambda^2 K_{\rm tr} - \zeta K_{\rm tr}\rVert_F$	1.20185×10^{-17}	Schur-complement match
$\lVert K_{\rm tr}(3K_{\rm tr}) - I\rVert_F$	3.84593×10^{-16}	inverse convention
$2\pi(17/178) - \arg\zeta$	4.14753×10^{-5} rad	side phase diagnostic
$\lvert\zeta_{178} - \zeta_{\rm tar}\rvert$	5.40970×10^{-6}	discrete approximation
$-\log\lvert\zeta\rvert$	2.0369040025655094	instanton-spurion option
$\lvert\lambda\rvert^2/(16\pi^2)$	8.25970×10^{-4}	canonical-basis audit scale
$\Delta b^X_{\rm vis}$	$(0, 0, 0)$	visible threshold silence

TABLE I. Optional Route-B algebraic verification.

Appendix C: Optional Route-B Algebraic Checks

This table checks only the optional Route-B Schur-complement algebra. It is not a substitute for the deferred full flavor, proton-decay, threshold, or source-sector audits.

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